

hospital2021

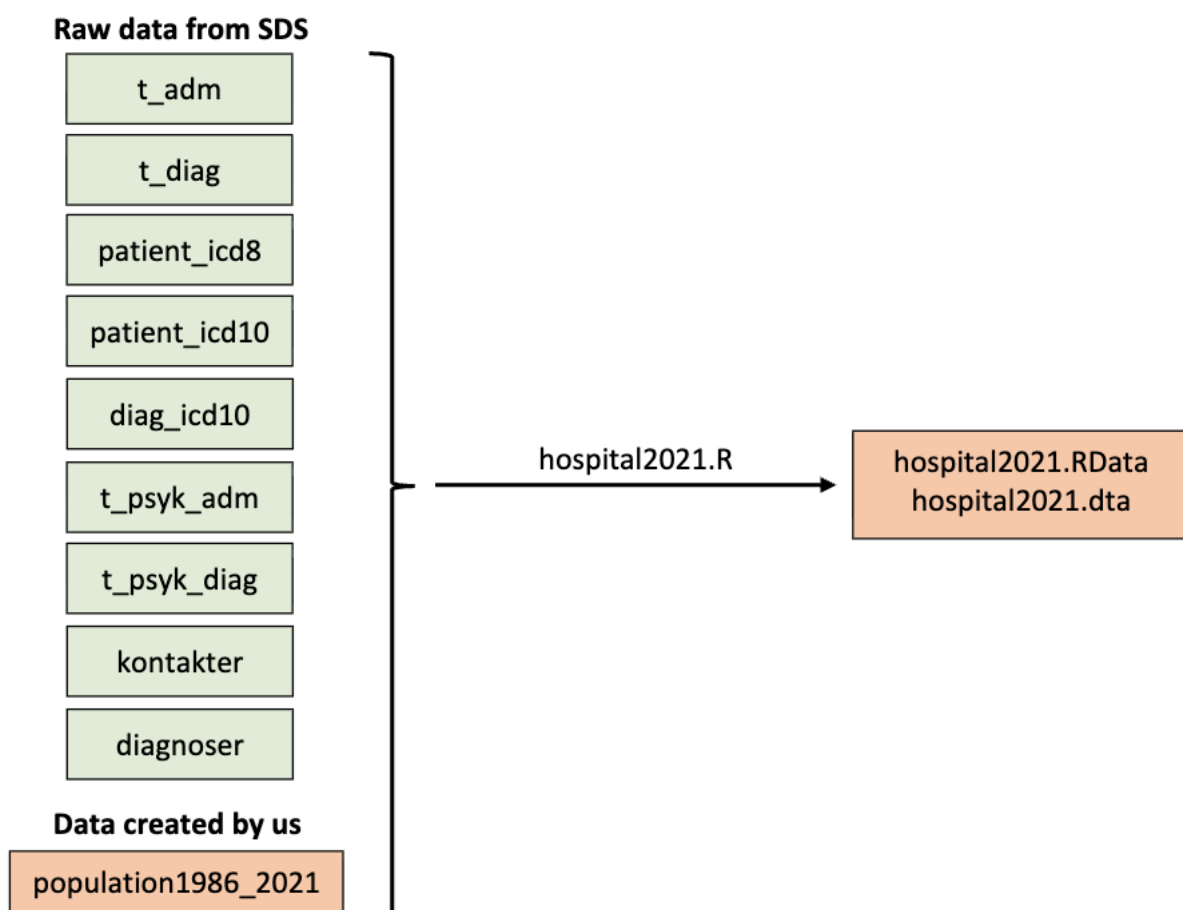
Version: 2 (September 30, 2024)

This document is available at Open Science Framework (OSF) (<https://osf.io/zhfyp/>). If you have any questions or suggestions, or you identify a mistake in our documentation or code, please contact us at opr@clin.au.dk.

Changes from previous version: we fixed a coding mistake that would classify around 0.8% of inpatient contacts as outpatient or emergency room visits. Those misclassified were based on LPR3 (years 2019-2022), and hospital contacts before 2019 were not affected by this mistake.

The dataset `hospital2021` contains information on all diagnoses in Danish hospitals (psychiatric and somatic departments) until August 24, 2022 for all individuals living in Denmark between January 1, 1986, and December 31, 2021. There is one observation for each diagnosis, including date of diagnosis, type of patient, and specific diagnosis code.

Code: hospital2021.R (R code)
Output: hospital2021.RData (R file)
hospital2021.dta (Stata file)



The dataset is created based on the following registers:

- Register `population1986_2021`, which includes basic information on all individuals living in Denmark between January 1, 1986, and December 31, 2021. We have created this dataset based on the registers from [Statistics Denmark \(DST\)](#) on

population ([‘Befolkning’; BEF](#)) and deaths in Denmark ([‘Døde i Danmark’; DOD](#)). Specific details and R code are available on OSF (<https://osf.io/zhfyp/>).

- The Danish National Patient Register¹ version LPR2 ([‘Landspatientregisteret’; LPR2](#)) from 1977 to March 2019 (datasets `t_adm` and `t_diag`), which contains information on admissions to inpatient facilities since 1977 and visits to outpatient and emergency departments since January 1, 1995. Hospital contacts in psychiatric units are not included in this register. The diagnostic system used is the Danish modification of the *International Classification of Diseases, Eighth Revision* (ICD-8) from 1977 to 1993, and *Tenth Revision* (ICD-10) from 1994 onwards.
- The Danish Psychiatric Central Research Register² from 1969 to March 2019, which contains information on admissions to psychiatric inpatient facilities since April 1969 and visits to outpatient psychiatric and emergency departments since January 1, 1995. From January 1, 1970, all psychiatric departments were committed to report the data, and it is the first year with valid data. The diagnostic system used is the ICD-8 from 1969 to 1993, and ICD-10 from 1994 onwards. The register is available through different specific registers from the [Danish Health Data Authority \(Sundhedsdatastyrelsen; SDS\)](#):
 - o The Danish Psychiatric Central Research Register ([‘Det Psykiatriske Centrale Forskningsregister’; PCR](#)) from 1969 to 1994 (dataset `patient_icd8` from 1969 to 1993; datasets `patient_icd10` and `diag_icd10` in 1994).
 - o The Danish National Patient Register¹ version LPR2 ([‘Landspatientregisteret’; LPR2](#)) from 1995 to March 2019 (datasets `t_psyk_adm` and `t_psyk_diag`).
- The Danish National Patient Register¹ version LPR3 ([‘Landspatientregisteret’; LPR3](#)) from February 2019 onwards (datasets `kontakter` and `diagnoser`). The diagnostic system used is the ICD-10. [According to SDS](#), there is a combination of LPR2 and LPR3 data between January and March 2019.

Dataset `hospital2021` contains all diagnoses classified as main diagnoses (*hoveddiagnoser/aktionsdiagnoser*; type ‘A’), auxiliary diagnoses (*bidiagnoser*; type ‘B’), basic diagnoses (*grundmorbus*; type ‘G’), and associated diagnoses (*tillægsdiagnoser*; type ‘+’). The dataset does not include referral diagnoses (*henvisningsdiagnose*; type ‘H’) or diagnoses linked to a hospital contact in which the main diagnosis is related to healthy individuals (i.e., ICD-8 code Y719; ICD-10 codes Z03.2, Z03.8, Z03.9, Z76.3).

The final dataset contains 401,001,287 observations from 8,046,713 unique identifiers with the following variables:

- **pnr**: Unique personal identifier. All individuals born or immigrating to Denmark receive a unique personal registration number (CPR number) containing 10 digits. Based on different digits’ positions, the CPR number contains information on the date of birth and sex. The [PNR number](#) is a de-identified version of the CPR number made by DST that does not include personal information but allows linkage between different registers based on this unique identifier.
- **sex**: Sex based on DST’s variable [koen](#) (1: male; 2: female), indicating the legal sex of the individual based on information from the CPR number. For more information, please look at documentation for register `population1986_2021` (available on OSF; <https://osf.io/zhfyp/>).

- **birth_d**: Date of birth based on DST's variable [foed_dag](#).
- **date_dx**: Date of diagnosis.
- **date_ending**: Date of discharge.^a
- **type_patient**: Type of patient (1: inpatient; 2: outpatient; 3: emergency room visit).^b
- **length_hours**: Length of the hospital visit in hours (only available for LPR3 to identify the type of patient).^b
- **diag**: ICD code for diagnosis (in ICD-8 or ICD-10).
- **type_diag**: Type of diagnosis (A: main; B: auxiliary; G: basic; +: associated).
- **recnum**: Identifier of hospital contacts.
- **lpr**: LPR version (2: LPR2; 3: LPR3).
- **hosp_type**: Origin of hospital contact (psychiatric, somatic).

^a According to [the Danish Health Data Authority guidelines](#), date of discharge for outpatients is registered on the final day for a course of treatments, and can occur months after first contact. Date of discharge for inpatients is when they leave hospital. If no date of discharge is registered for emergency room visits, date of discharge should be set to date of first contact (date of diagnosis).

^b While the LPR2 (used until March 2019) has a clear definition of type of patient, this is not the case for LPR3. For LPR3, we followed the guidelines on how to classify patients according to the [Danish Health Data Authority suggestions](#) and a previous publication comparing LPR2 with LPR3 data.³ In brief, patients were classified as 'inpatients' if they stayed at the hospital for 8 hours or longer; 'emergency room visits' if they were at the hospital for less than 8 hours and they were attended in the emergency unit (*skadestue*) and priority set to acute patients (*ATA1*); or 'outpatient' otherwise. The variable 'length_hours' can be used to easily change the definition between inpatients and emergency room visits.

1. Lynge E, Sandegaard JL, Rebolj M. The Danish National Patient Register. *Scandinavian Journal of Public Health*. 2011;39(7 Suppl):30-33. doi:10.1177/1403494811401482
2. Mors O, Perto GP, Mortensen PB. The Danish Psychiatric Central Research Register. *Scand J Public Health*. 2011;39(7 Suppl):54-57. doi:10.1177/1403494810395825
3. Bernstorff M, Hansen L, Perfalk E, Danielsen AA, Østergaard SD. Stability of diagnostic coding of psychiatric outpatient visits across the transition from the second to the third version of the Danish National Patient Registry. *Acta Psychiatrica Scandinavica*. 2022;146(3):272-283. doi:10.1111/acps.13463
4. Pedersen CB, Mors O, Bertelsen A, et al. A comprehensive nationwide study of the incidence rate and lifetime risk for treated mental disorders. *JAMA Psychiatry*. 2014;71(5):573-581. doi:10.1001/jamapsychiatry.2014.16

